

Business Requirement Document

Notification Service Module

ShopSphere E-Commerce Platform — ShopSphere Platform Modernization Program (FY2026)

Document Title	Business Requirement Document — Notification Service Module
Document ID	BRD-EC-NTS-010
Version	1.0
Author	Neha Kapoor, Business Analyst — Customer Engagement
Date	19-Mar-2026
Status	Approved

1. Document Information

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2. Executive Summary

ShopSphere Technologies Pvt. Ltd. operates the ShopSphere E-Commerce Platform, a B2C marketplace serving approximately 2 million monthly active users across India, UAE, and Southeast Asia. Customer engagement across the purchase lifecycle — from registration through order fulfillment and post-purchase support — depends on timely, reliable, and relevant communication delivered across multiple channels. The Notification Service Module is being commissioned under the ShopSphere Platform Modernization Program (FY2026) to replace fragmented, point-to-point notification logic embedded in individual application modules with a centralized, event-driven notification platform.

This document defines the business requirements for a Notification Service capable of dispatching transactional notifications (registration confirmation, OTP, order confirmation, shipping updates, payment receipts, return/refund status) and marketing notifications (opt-in promotional campaigns, abandoned cart reminders) across email, SMS, push, and in-app banner channels. The service will provide a customer-facing preference center, centralized and localized template management, and channel fallback logic to maximize delivery reliability while respecting customer consent and regulatory obligations under applicable data privacy laws.

3. Business Objective

The primary business objective is to establish a single, reliable, and compliant notification backbone that increases customer trust and engagement while reducing operational overhead associated with maintaining duplicate notification logic across the Registration, Checkout, Order Management, Returns & Refunds, and Customer Reviews modules. By consolidating notification dispatch into a dedicated service, ShopSphere aims to: improve transactional notification delivery reliability to reduce customer support inquiries related to missed order updates; increase marketing campaign effectiveness through better-targeted, opt-in communications and reduced notification fatigue; ensure consistent multi-language customer communication across English, Hindi, and Arabic; and achieve full compliance with regional data privacy and consent regulations, including the India DPDP Act and UAE data protection law, thereby reducing regulatory and reputational risk.

4. Project Scope

In Scope

- Transactional notifications: registration confirmation, OTP delivery, order confirmation, shipping and delivery updates, payment receipts, and return/refund status updates.
- Marketing notifications: opt-in promotional campaigns and abandoned cart reminder notifications.
- Multi-channel delivery: email, SMS, push notification, and in-app banner.
- Customer-facing notification preference center allowing granular opt-in/opt-out control by channel and notification category.
- Centralized template management with versioning and localization support for English, Hindi, and Arabic.
- Channel fallback logic to reroute failed deliveries to an alternate channel.
- Rate limiting and throttling controls to prevent notification fatigue.
- Delivery status tracking, retry/backoff handling, and audit logging of all dispatched notifications.

Out of Scope

- Outbound voice call / Interactive Voice Response (IVR) notifications.
- WhatsApp Business API integration (planned as a future enhancement).
- AI-based send-time optimization and personalization engine.

5. Stakeholders

Name	Role	Department	Responsibility
Rajesh Mehta	VP of E-Commerce	Business	Executive Sponsor
Ananya Sharma	Product Owner, ShopSphere Platform	Product	Prioritization & sign-off
Neha Kapoor	Business Analyst, Customer Engagement	Business Analysis	Requirements owner
Sarah Chen	Enterprise IT Architect	IT/Architecture	Technical feasibility
Michael Torres	QA Lead	Quality Assurance	Test strategy & sign-off
David Kim	DevOps Lead	Engineering/Infra	Deployment & monitoring
Fatima Al-Sayed	Compliance & Security Officer	Risk & Compliance	Data privacy & opt-in consent

6. Business Requirements

ID	Business Requirement
BR-001	The business requires a centralized notification service so that all transactional and marketing customer communications originate from a single, auditable platform instead of fragmented module-level logic.
BR-002	The business requires reliable, at-least-once delivery of transactional notifications so that customers receive timely confirmation of critical order and account events, reducing support escalations.
BR-003	The business requires multi-channel delivery with automatic fallback so that customers still receive time-sensitive notifications (such as OTP and shipping updates) even when a preferred channel is unavailable.
BR-004	The business requires a self-service notification preference center so that customers can control the channels and categories of communication they receive, supporting consent compliance and reducing opt-out rates.
BR-005	The business requires centralized, versioned, and localized notification templates so that messaging remains consistent, brand-compliant, and available in English, Hindi, and Arabic across all markets.
BR-006	The business requires the notification service to scale to handle peak sale event traffic so that flash sale and promotional campaigns can reach millions of customers without service degradation.
BR-007	The business requires rate limiting and frequency capping on marketing notifications so that customer notification fatigue is minimized and unsubscribe rates are kept within acceptable thresholds.
BR-008	The business requires full compliance with regional data privacy and consent regulations so that ShopSphere avoids regulatory penalties and reputational damage in India, UAE, and Southeast Asian markets.

7. Functional Requirements

Transactional Notifications

- **FR-001** The system shall generate and dispatch a registration confirmation notification via email and, where a valid mobile number is provided, SMS, immediately upon successful account creation.
- **FR-002** The system shall generate and dispatch OTP notifications via SMS (primary) with email fallback for account verification, login, and checkout authentication events within the performance SLA.
- **FR-003** The system shall generate an order confirmation notification across email and push channels immediately upon receipt of an order-placed event from the Order Management module, including order number, SKU list, and estimated delivery date.
- **FR-004** The system shall dispatch shipping status update notifications (dispatched, in-transit, out-for-delivery, delivered) triggered by carrier status events, using push as the primary channel with SMS fallback.
- **FR-005** The system shall generate a payment receipt notification via email upon confirmation of successful payment capture, including itemized charges and applicable taxes.
- **FR-006** The system shall dispatch return and refund status notifications (RMA initiated, item received, refund processed) triggered by events from the Returns & Refunds module across email and in-app banner channels.

Marketing Notifications

- **FR-007** The system shall support the creation and scheduling of opt-in promotional campaign notifications targeted by customer segment, channel, and region.
- **FR-008** The system shall trigger abandoned cart reminder notifications when a cart remains unconverted for a configurable idle period (default 4 hours), subject to the customer's marketing consent status.
- **FR-009** The system shall suppress marketing notifications to any customer who has opted out of the relevant channel or category, honoring suppression lists synced from Salesforce CRM.

Preference Center

- **FR-010** The system shall provide a customer-facing preference center allowing customers to opt in or out of notification channels (email, SMS, push, in-app) independently for transactional and marketing categories, subject to regulatory minimums for mandatory transactional alerts.
- **FR-011** The system shall synchronize customer preference changes with Salesforce CRM in near real time so that all customer-facing teams have a consistent view of communication consent.

Template Management

- **FR-012** The system shall provide a centralized template management interface enabling authorized business users to create, edit, version, and publish notification templates without requiring code deployment.
- **FR-013** The system shall support localized template variants in English, Hindi, and Arabic, with automatic selection based on customer locale preference.
- **FR-014** The system shall apply channel fallback logic such that if delivery on the primary channel fails or is unconfirmed within a configurable timeout, the notification is automatically re-attempted on the next configured fallback channel (e.g., SMS if push fails).
- **FR-015** The system shall log the delivery status (queued, sent, delivered, failed, bounced) of every dispatched notification and expose this status via API for downstream reporting and support tooling.

8. Non-Functional Requirements

Performance

- **NFR-001** Transactional notifications shall be dispatched within 30 seconds of the triggering event at the 95th percentile.
- **NFR-002** The template rendering and localization engine shall resolve the correct localized template within 200 milliseconds per request.

Security

- **NFR-003** The system shall comply with OWASP Top 10 controls for all customer-facing and API interfaces.
- **NFR-004** All PII, including customer contact details and preference records, shall be encrypted using AES-256 at rest and TLS 1.2+ in transit.

- **NFR-005** Consent and opt-in/opt-out records shall be maintained in accordance with the India DPDP Act and UAE data protection law, with a full audit trail of consent state changes.

Availability

- **NFR-006** The notification dispatch service shall maintain 99.9% uptime, measured monthly.
- **NFR-007** Planned maintenance windows shall not exceed 4 hours per calendar quarter and shall be scheduled outside peak sale events.

Scalability

- **NFR-008** The system shall support burst notification volume of up to 15 times normal daily volume during flash sale and peak promotional events without degradation of transactional SLA.
- **NFR-009** The system architecture shall support horizontal scaling of dispatch workers per channel independently.

Reliability

- **NFR-010** The system shall guarantee at-least-once delivery semantics with automated retry and exponential backoff for failed sends, up to a configurable maximum retry count.
- **NFR-011** Recovery Point Objective (RPO) shall not exceed 15 minutes and Recovery Time Objective (RTO) shall not exceed 1 hour for the notification service in the event of a disaster recovery failover.

9. User Stories

- **US-001** As a customer, I want to receive an OTP via SMS during checkout, so that I can securely verify my identity and complete my purchase.
- **US-002** As a customer, I want to receive shipping status updates via push notification, so that I know exactly when my order will arrive.
- **US-003** As a customer, I want to manage my notification preferences in one place, so that I only receive communications I find relevant and useful.
- **US-004** As a customer, I want to receive an abandoned cart reminder if I opted in to marketing messages, so that I don't forget items I intended to purchase.
- **US-005** As a marketing manager, I want to schedule promotional campaigns targeted to opted-in customer segments, so that I can drive engagement without violating consent rules.
- **US-006** As a customer support agent, I want to view the delivery status history of a customer's notifications, so that I can quickly resolve disputes about missed order updates.
- **US-007** As a content manager, I want to edit and publish localized notification templates without engineering support, so that I can update messaging quickly across markets.
- **US-008** As a compliance officer, I want an audit trail of all consent changes, so that I can demonstrate regulatory compliance during an audit.

10. Acceptance Criteria

ID	Related ID	Criteria	Expected Result
AC-001	FR-002	Given a customer initiates checkout login, when an OTP is requested, then the SMS shall be dispatched within 30 seconds.	OTP received on registered mobile number within SLA.
AC-002	FR-003	Given an order is successfully placed, when the order-placed event is received, then an order confirmation notification is sent via email and push.	Customer receives order confirmation on both channels.
AC-003	FR-014	Given a push notification fails to deliver within the configured timeout, when the fallback rule is evaluated, then the notification is re-sent via SMS.	Customer receives the notification via SMS fallback.
AC-004	FR-010	Given a customer opts out of marketing SMS in the preference center, when a promotional campaign is triggered, then the customer is excluded from the SMS send list.	No marketing SMS sent to opted-out customer.
AC-005	FR-008	Given a customer's cart remains idle for 4 hours, when the abandonment threshold is reached and the customer has marketing consent, then an abandoned cart reminder is triggered.	Reminder notification dispatched to consented customer only.

AC-006	FR-013	Given a customer's locale is set to Arabic, when a transactional notification is generated, then the Arabic-localized template variant is used.	Notification content renders correctly in Arabic.
AC-007	US-006	Given a support agent looks up a customer's order, when they view notification history, then all dispatch attempts and statuses are visible.	Full delivery status history displayed to agent.
AC-008	NFR-010	Given a notification dispatch attempt fails, when the retry policy is invoked, then the system retries with exponential backoff up to the configured maximum.	Notification eventually delivered or marked permanently failed after max retries.
AC-009	FR-009	Given a customer is on the CRM suppression list, when a marketing campaign is dispatched, then that customer is excluded from all campaign sends.	Suppressed customer receives no marketing communication.

11. Business Rules

- **RULE-001** Mandatory transactional notifications (OTP, payment receipt, order confirmation) cannot be fully opted out of by the customer, though channel preference within available channels may be adjusted.
- **RULE-002** Marketing notifications shall only be sent to customers with an active, verifiable opt-in consent record.
- **RULE-003** A maximum of 3 marketing notifications per customer per day shall be enforced across all channels combined to prevent notification fatigue.
- **RULE-004** Channel fallback shall be attempted a maximum of one time per notification, and only for notifications marked as time-sensitive.
- **RULE-005** All notification templates must be approved and published through the template management workflow before use in production.
- **RULE-006** Consent state changes must be reflected in both the Notification Service and Salesforce CRM within 5 minutes of the change.
- **RULE-007** Notifications that fail delivery after the maximum retry count shall be logged as permanently failed and flagged for customer support review if the notification category is transactional.

12. Process Flow Description

Primary Flow — Order Shipped Notification:

1. The Order Management module publishes an "order shipped" event containing order ID, customer ID, carrier, and tracking number.
2. The Notification Service consumes the event from the event feed and identifies the customer's applicable notification preferences and locale from the Preference Center and Salesforce CRM.
3. The service selects the appropriate localized template for the "shipping update" category.
4. The service determines the primary channel (push, via Firebase Cloud Messaging) based on customer preference and channel availability.
5. The rendered notification is dispatched to Firebase Cloud Messaging for delivery to the customer's mobile app.
6. The delivery status (sent/delivered/failed) is recorded against the notification record.
7. If delivery is confirmed, the workflow completes and the status is logged as "delivered".

Alternate Flow — Primary Channel Delivery Failure:

1. If the push notification is not confirmed as delivered within the configured timeout (default 60 seconds), the system evaluates the fallback rule for time-sensitive notifications.
2. The system selects the configured fallback channel (SMS via Twilio SMS Gateway) and re-renders the notification using the SMS-localized template.
3. The fallback notification is dispatched via Twilio SMS Gateway to the customer's registered mobile number.
4. The delivery status of the fallback attempt is recorded, and the original push attempt is marked "failed – fallback triggered".
5. If the fallback also fails, the notification is logged as permanently failed and flagged for customer support review per RULE-007.

13. Assumptions

- Customer contact details (email, mobile number) captured during registration are accurate and validated at the point of capture.

- Salesforce CRM remains the system of record for customer communication preferences and suppression lists.
- SendGrid Email API, Twilio SMS Gateway, and Firebase Cloud Messaging accounts are provisioned with sufficient sending capacity and reputation standing prior to go-live.
- Source modules (Registration, Checkout, Order Management, Returns & Refunds, Customer Reviews) will publish standardized events to the notification event feed.
- Localization content (Hindi and Arabic translations) will be provided by the marketing content team ahead of template publishing.
- Peak sale event volumes are forecast and communicated to the DevOps team at least 2 weeks in advance for capacity planning.

14. Constraints

- The Notification Service must operate within existing SendGrid, Twilio, and Firebase Cloud Messaging vendor contracts and rate limits during the initial rollout.
- WhatsApp Business API and voice/IVR channels are explicitly excluded from this phase and cannot be substituted for SMS or push without a scope change.
- All customer data handling must comply with the India DPDP Act and UAE data protection law, limiting cross-border data transfer options.
- Template management must not require engineering deployment cycles, constraining the solution to a configuration-driven or low-code approach.
- The program budget for FY2026 limits initial regional rollout to India, UAE, and Southeast Asia only.

15. Dependencies

- Registration module must emit account-creation events to trigger registration confirmation and OTP notifications.
- Checkout and Order Management modules must emit order-placed, payment-captured, and shipping-status events in a standardized event schema.
- Returns & Refunds module must emit RMA and refund status events.
- Salesforce CRM must expose an API for real-time synchronization of customer preferences and suppression lists.
- SendGrid Email API, Twilio SMS Gateway, and Firebase Cloud Messaging must be available and integrated prior to go-live.

16. Risks

Risk ID	Description	Likelihood	Impact	Mitigation
RISK-001	Third-party channel providers (SendGrid, Twilio, FCM) experience outages during peak sale events.	Medium	High	Implement channel fallback logic and monitor provider status dashboards with automated alerting.
RISK-002	Inaccurate or stale consent data leads to non-compliant marketing sends.	Medium	High	Enforce near real-time sync with Salesforce CRM and periodic consent audits.
RISK-003	Notification fatigue results in elevated opt-out and unsubscribe rates.	Medium	Medium	Apply rate limiting and frequency capping per RULE-003.
RISK-004	Burst traffic during flash sales exceeds provisioned dispatch capacity.	Medium	High	Load-test for 15x normal volume and implement auto-scaling of dispatch workers.
RISK-005	Incorrect localization leads to customer confusion or brand damage in non-English markets.	Low	Medium	Mandatory translation review workflow prior to template publishing.
RISK-006	Delayed event publishing from source modules causes notification SLA breaches.	Low	Medium	Define and monitor event-publishing SLAs with source module owners.

17. Data Requirements

The Notification Service shall capture and retain notification event, delivery, and consent data to support operational monitoring, customer support, and compliance reporting. Notification records shall be retained for a minimum of 24

months for transactional notifications and 12 months for marketing notifications, after which they shall be archived per the company data retention policy.

Data Field	Description
Notification ID	Unique identifier for each notification instance.
Customer ID	Reference to the customer record; linked to CRM profile.
Channel	Delivery channel used (email, SMS, push, in-app banner).
Notification Category	Classification (transactional or marketing) and sub-type (e.g., OTP, shipping update).
Template ID / Version	Reference to the template and version used for rendering.
Locale	Language variant applied (English, Hindi, Arabic).
Delivery Status	Current status: queued, sent, delivered, failed, bounced.
Consent Flag	Indicator of active opt-in/opt-out status at time of send.
Timestamp	Event trigger time and each status transition time.

18. Integration Requirements

System	Purpose	Direction	Data Exchanged
SendGrid Email API	Dispatch transactional and marketing email notifications.	Outbound	Rendered email content, recipient address, delivery status webhook.
Twilio SMS Gateway	Dispatch OTP and transactional/marketing SMS.	Outbound	Message body, recipient mobile number, delivery status callback.
Firebase Cloud Messaging (FCM)	Deliver mobile app push notifications and in-app messages.	Outbound	Push payload, device token, delivery/read receipt.
Salesforce CRM	Sync customer notification preferences and suppression lists.	Bidirectional	Consent status, opt-out flags, communication history.
Order Management / Checkout / Returns & Refunds / Registration / Customer Reviews	Provide event triggers for transactional notifications.	Inbound	Order, payment, shipment, RMA, and account event payloads.

19. Reporting Requirements

- Daily notification volume dashboard segmented by channel and category (transactional vs. marketing).
- Delivery success/failure rate report by channel, with drill-down to failure reason.
- Opt-out and unsubscribe trend report by category and region.
- Average dispatch latency report against the 30-second SLA target.
- Consent audit report showing consent state change history for compliance reviews.
- Peak-event load report comparing actual burst volume against 15x baseline capacity planning.

20. Success Metrics

Metric	Target
Transactional notification delivery success rate	≥ 99.5%
Average dispatch latency (95th percentile)	≤ 30 seconds
Notification service uptime	≥ 99.9%
Marketing opt-out rate	≤ 2% per campaign
Channel fallback success rate	≥ 95%
Reduction in support tickets for missed order notifications	≥ 30% within 2 quarters post-launch

21. Future Enhancements

- Integration with WhatsApp Business API as an additional notification channel.
- Outbound voice call / IVR notifications for critical account and fraud alerts.
- AI-based send-time optimization and personalization engine to improve engagement rates.
- Expanded localization support for additional Southeast Asian languages beyond English, Hindi, and Arabic.

- Advanced customer segmentation and A/B testing capabilities for marketing campaigns.

22. Glossary

Term	Definition
SKU	Stock Keeping Unit; a unique identifier for a distinct product/item.
PDP	Product Detail Page; the page displaying detailed information about a single product.
PLP	Product Listing Page; the page displaying a list of products, typically by category or search.
Cart Abandonment	When a customer adds items to a shopping cart but does not complete the purchase.
COD	Cash on Delivery; a payment method where payment is made upon delivery of goods.
RMA	Return Merchandise Authorization; an authorization for a customer to return a purchased item.
SLA	Service Level Agreement; a defined commitment for a service metric such as latency or uptime.
OTP	One-Time Password; a temporary code used for identity verification.
KYC	Know Your Customer; identity verification processes required for regulatory compliance.
GMV	Gross Merchandise Value; total value of merchandise sold over a given period.
PII	Personally Identifiable Information; data that can identify a specific individual.
Fallback Channel	An alternate delivery channel used automatically when the primary channel fails.

23. Appendix

Reference Documents: ShopSphere Platform Modernization Program Charter (FY2026); ShopSphere Data Privacy & Consent Management Policy; Salesforce CRM Integration Specification.

Related BRDs: BRD-EC-REG-001 Registration Module, BRD-EC-CHK-004 Checkout Module, BRD-EC-ORD-006 Order Management Module, BRD-EC-RET-008 Returns & Refunds Module, BRD-EC-REV-009 Customer Reviews Module.

Version	Date	Author	Change Description
0.1	02-Feb-2026	Neha Kapoor	Initial draft for stakeholder review.
0.9	05-Mar-2026	Neha Kapoor	Incorporated feedback from Architecture and Compliance review.
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